CS 839 Programming Assignment
Sustained Two-Leg Support for MuJoCo Ant

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1 Interesting Behavior

The Ant should remain upright with exactly two feet on the ground for as much of each episode as possible.

2 Task MDP

State.
A 118-dimensional vector: the standard Ant joint, velocity, and contact-force observation, plus four foot-contact indicators, normalized target hold progress, and a two-element target-pair indicator.

Action.
Eight continuous joint torques in $[-1, 1]^8$.

Reward.
Let I_t mark a valid two-leg pose, h_t and u_t be clipped height and upright scores, d_t target-contact quality, p_t hold progress, b_t a completed target hold, B_t body contact, and D_t termination:

$$r_t = c_t + h_t + u_t + 4I_t + 0.5d_t h_t u_t + 2p_t + 2b_t - 4B_t - 0.02\|\omega_t\|^2 - 0.02\|a_t\|^2 - 20D_t,$$

where $c_t = \max(0, 1 - 0.5|n_t - 2|)$ and n_t is the number of grounded feet.

Episode.
1,000 steps at 0.05 seconds per step, or 50 seconds.

Termination.
Terminate if torso up-alignment is below -0.2 or the simulation becomes non-finite; otherwise truncate at 1,000 steps.

The observation includes the mechanical state and reward-relevant task memory, so I treat it as Markov for this position-invariant task.

3 True Performance Metric

A valid pose has exactly two grounded feet, torso height at least 0.45 m, positive torso up-alignment, and no other body contact. Performance is

$$M = 0.05 \sum_{t=1}^{T} \mathbf{1}[\text{valid two-leg pose at } t],$$

measured in seconds from 0 to 50. It directly measures the intended behavior and excludes reward-shaping terms.

4 Training Performance

I used paired training seeds 1–5. I smoothed each seed with a 20-episode rolling mean, interpolated it onto a 5,000-step grid, and then averaged. Thin lines are individual seeds; shading is the 95% Student-t interval $\bar{x} \pm 2.776 s/\sqrt{5}$.

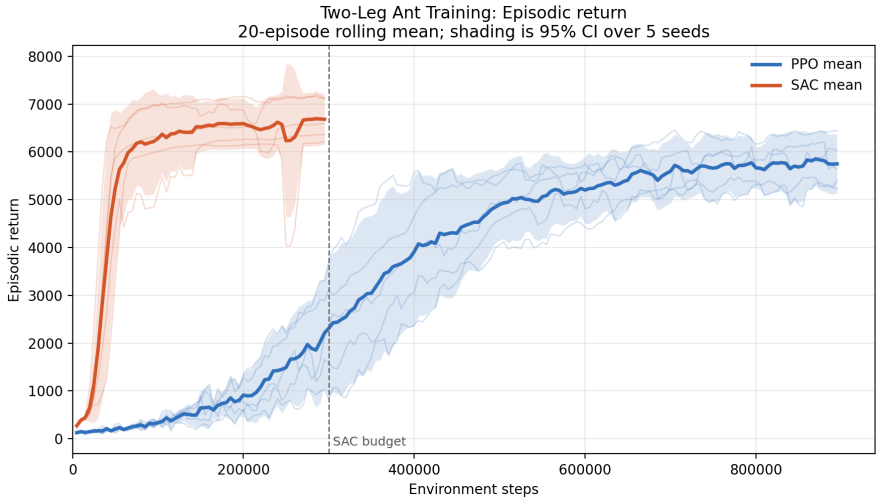


Figure 1: Mean episodic return. SAC ends at 300,000 steps; PPO continues to approximately 899,000 steps.

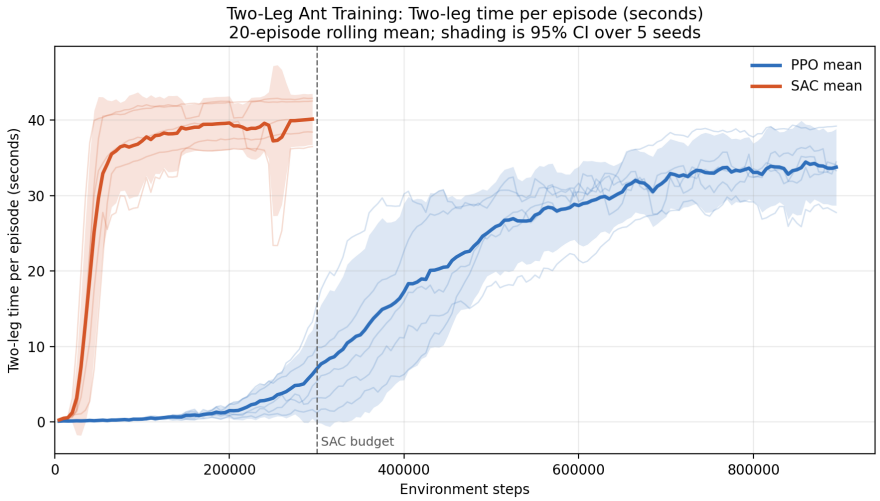


Figure 2: Mean true performance: valid two-leg seconds per 50-second episode.

5 Algorithm Comparison

Final deterministic evaluation used the same ten evaluation seeds for every policy. Values are means and 95% intervals across five training seeds.

	Return	Two-leg time (s)	Fraction	Longest hold (s)
PPO	6011 ± 386	36.11 ± 3.58	0.722 ± 0.072	0.969 ± 0.539
SAC	6777 ± 571	40.66 ± 3.09	0.813 ± 0.062	1.111 ± 0.287

Final performance. SAC achieved the higher mean true performance: 40.66 two-leg seconds per episode, compared with 36.11 for PPO. SAC learned substantially faster. Its smoothed mean exceeded 35 two-leg seconds near 65,000 steps, whereas PPO did not exceed 30 seconds until about 645,000 steps. This is likely because SAC is off-policy and reuses transitions from its replay buffer, while PPO is on-policy and discards data after each update.

Variance. SAC was slightly more consistent on the true metric. Its across-seed standard deviation was 2.49 seconds, compared with 2.89 seconds for PPO. However, SAC had greater variation in episodic return, with standard deviations of 460 for SAC and 311 for PPO.

Tuning and limitations. I first tried rewards emphasizing survival, torso height, and a sparse success bonus. These produced upright movement but did not reliably maintain exactly two contacts. I also tried rewarding alternating diagonal pairs, but the policies usually selected one easier pair and recorded no valid switches. The final reward increased the dense two-leg-pose reward and added hold-progress shaping. This improved total two-leg time, but the longest continuous holds were only about one second. Thus the policies learned aggregate two-leg contact, but not stable alternating support.

6 Best Policy Video

`best_policy_sac_seed3.mp4` shows the best policy: SAC seed 3, averaging 43.12 two-leg seconds (86.24% of an episode). It stays upright but does not alternate support pairs.

7 Reproducibility

Code is included in the zip. Exact paired-run commands are:

```
for seed in 1 2 3 4 5; do
  .venv/bin/python cleanrl/ppo_two_leg.py --env-id AntTwoLeg-v0 \
    --exp-name ppo_sustained_hold_v4 --seed "$seed" \
    --total-timesteps 900000 --eval-episodes 10 --capture-video --save-model
done
for seed in 1 2 3 4 5; do
  .venv/bin/python cleanrl/sac_two_leg.py --env-id AntTwoLeg-v0 \
    --exp-name sac_sustained_hold_v4 --seed "$seed" \
    --total-timesteps 300000 --learning-starts 5000 --buffer-size 300000 \
    --eval-episodes 10 --capture-video --save-model
done
.venv/bin/python analysis/two_leg_five_seed/analyze_runs.py
```

AI Use Disclosure

I used Codex for partial code implementation, debugging and report checking.